

Observation Recency in Bernoulli Fusion for Intermittent Multirobot Tracking

Abstract—Intermittent contact between mobile robots brings together target beliefs with different observation histories. Weighting an entire Bernoulli density by direct-observation age can improve existence decisions while degrading spatial pooling. We study a lightweight alternative in an existing labeled multi-Bernoulli tracker: qualify observation lineage, retain ordinary spatial weights, and apply direct-opportunity recency only to existence terms. Recent missed detections remain valid evidence, and relays do not refresh local direct age. A two-block variational formulation establishes the fixed-input spatial invariance of this rule, without claiming recursive accuracy or cardinality consistency. In three prescribed eight-robot simulation families with twenty new paired seeds each, the rule reduces mean OSPA by 3.1% and 24.7% against a lineage-aware control in split-rejoin and churn-departure scenarios. Common-target localization RMSE remains within 0.02% of the control, whereas applying age to both blocks increases it by 16.8% and 3.1%. False-target behavior remains a limitation: the departure benefit seen during development does not reproduce, and confirmation trades fewer false hypotheses for slower discovery. The findings concern the placement of temporal trust in posterior fusion under shared birth labels and known poses, not a complete robotic perception or navigation system.

I. Introduction

Mobile robots often acquire complementary views while moving between connected and disconnected regions. When two subteams meet again, a newly received track can be supported by recent local measurements, inherited from an earlier encounter, or still close to an unobserved birth prior. Conversely, a low existence probability can reflect a useful recent missed detection or merely a lack of sensing opportunity. These distinctions matter when the communication opportunity permits only a small number of fusion rounds. A geometrically valid link does not make the information at its two ends equally current. Figure 1 shows the prescribed contact patterns studied here.

Random finite set filters provide a natural representation of uncertain target number and state. The labeled multi-Bernoulli (LMB) filter represents each candidate track by an existence probability and a conditional spatial density [1]. Geometric pooling, also called generalized covariance intersection or Kullback–Leibler averaging (KLA), is widely used when cross-correlations are unavailable [2]. Its interaction with unequal fields of view (FoVs), label support, and cardinality estimation is already well established. The practical question here is more specific: where should direct-observation recency enter a Bernoulli fusion rule when robots reunite?

Multiplying a source’s entire fusion weight by an age factor seems natural, but changes both existence and position pooling. A source that has not observed a track

recently can still carry a useful spatial estimate received from another robot. Reducing its spatial influence solely because its own sensor has been uninformative can therefore sacrifice localization while improving target-number estimates. We study an existence-only alternative: retain ordinary spatial weights on eligible inputs, and use direct-opportunity age to weight the Bernoulli existence terms. The spatial overlap normalizer is retained; this is not a claim of fully decoupled cardinality fusion.

We make three contributions within this restricted setting. First, we formulate an observation-history rule that distinguishes untouched priors, retained observation lineage, recent positive or negative opportunities, and observable label absence. Relaying does not refresh a robot’s direct age. Second, a two-block variational formulation gives the resulting Bernoulli pool and establishes a fixed-input spatial invariance property. Third, a paired simulation study separates discovery, departure, and false-positive effects under identical motion and packet schedules, including failed confirmation and both-block recency ablations.

The study changes only the fusion layer of an existing LMB implementation. The robot paths are prescribed and the inputs are synthetic point detections; there is no navigation, perception, or hardware claim. Birth regions and clocks are shared, and the label keys are already consistent. These assumptions isolate fusion behavior but limit the deployment scope. The development screens and their failures are retained, and the subsequent validation uses frozen parameters and new paired seeds.

II. Related Work

A. Cooperative robotic search and tracking

Dames studies distributed search and tracking with PHD filtering and robot motion [3]. Ramachandran et al. reconfigure a multirobot communication network after sensing-resource degradation, using a central optimization system [4]. DecSTER combines decentralized active search and tracking with unreliable, asynchronous inter-agent communication [5]. These works address decisions about where robots should move or how resources should be connected. We hold motion and attempted links fixed to isolate the effect of posterior fusion during intermittent contact.

B. Unequal views and information weighting

Wang et al. already use information divergence to adapt LMB fusion weights across multiple views [6]. FoV compensation and multi-view geometric fusion are also

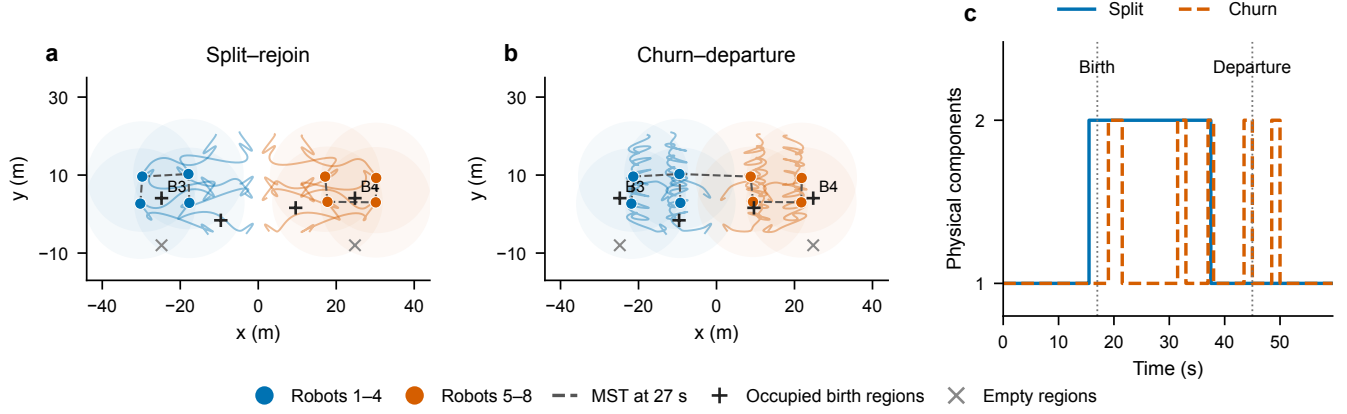


Fig. 1. Prescribed mobile sensing geometry (validation seed 2901, selected by index). Thin curves are complete robot paths; circles and dashed tree edges show the state at 27 s. Light disks denote 14 m FoVs. Crosses mark fixed candidate birth regions; the two lower regions are empty. The split and churn families produce different physical contact histories. Only the churn family includes the target departure at 45 s.

established approaches [7], [8]. Gao et al. derive LMB-constrained minimum-information-loss (MIL) pooling and decompose label spaces for unequal views [9], [10]. Li et al. explicitly recognize that retained or relayed information can remain useful outside the current nominal FoV, and develop a density-decomposition CPHD approach [11]. Jin et al. examine shared-prior label consistency and introduce a consensus information selector [12]. Communication-aware adaptive KLA/LMB weights also predate this study [13]. Thus neither variable per-track weights nor historical FoV information is claimed as new. We examine the narrower placement of a causal direct-age factor within an existing Bernoulli fusion recursion.

C. Existence and spatial fusion

Üney et al. show that pointwise consistency of finite-set geometric mixtures does not ensure cardinality consistency, and advocate separate cardinality and localization fusion [14]. Our use of different weight vectors is not a new general decoupling principle. Unlike fully independent cardinality pooling, the proposed expression retains the spatial overlap penalty. The contribution is the observation-history instantiation, its limited invariance property, and its measured tradeoffs at robot encounters. Label assignment remains a separate problem [15]; the shared-label MIL controls below are specializations, not full reproductions of independent-label systems.

III. Problem and Fusion Rule

A. Robot, sensing, and communication model

There are N robots with known poses $s_i(t)$ and a common clock. A physical edge exists when two robots are within radio range. A geometry-only coordinator selects a bidirectional minimum spanning tree in each connected component. At each time, a robot predicts its LMB, updates with its own measurement set, and

exchanges the resulting posterior for one synchronous fusion round. A dropped neighbor's Metropolis weight returns to the receiver's self weight. Fusion output is retained as the prior for the next local step. The coordinator is modeled as reliable and its traffic is charged; the estimator messages are lossy. This is not a decentralized tree-agreement protocol or a consensus-convergence guarantee.

A label $\ell = (t_b, b)$ specifies a shared birth time and candidate region. For represented label ℓ , source j supplies $\pi_j^\ell = (r_j^\ell, p_j^\ell)$, where $r_j^\ell \in [0, 1]$ and $p_j^\ell(x)$ is a Gaussian-mixture density over position and velocity. The birth prior does not disclose whether a target actually exists. All methods receive the same region/time prior and already-consistent keys. They do not use truth cardinality, target trajectories, or evaluation assignments in fusion decisions.

B. Observation history and participation

Each retained Bernoulli carries two metadata values. The lineage flag h_j^ℓ records whether the density descends from any scheduled local update with nonzero detection opportunity. A direct opportunity is evaluated on the predicted Gaussian means before the measurement update:

$$o_j^\ell(t) = \mathbf{1} \left[\sum_g w_{jg}^\ell p_{D,j}(\mu_{jg}^\ell, t) > 0 \right]. \quad (1)$$

The local timestamp τ_j^ℓ is the last time this condition held at robot j . Both detections and missed detections count. A received posterior can propagate lineage but cannot advance τ_j^ℓ : after fusion the receiver attaches its own local timestamp. Consequently the age measures direct sensor opportunity, not the age of every datum in a correlated posterior. We deliberately do not infer measurement novelty from packet arrival or attempt to remove all common information.

The participation rule is a controlled baseline component. If at least one input carries observation lineage, represented inputs without lineage are excluded as untouched priors. Otherwise they retain ordinary participation. An absent label contributes a negative existence censor only when the represented spatial density places at least half its Gaussian-mean mass inside that source's current FoV, with expected detection at least 0.1. Its existence upper bound is 0.01. Other absent sources abstain. Such censors are qualified negative evidence and are never assigned a spatial density. The flags propagate only through actual participating inputs. This rule uses a mean-based FoV approximation, not exact integration through a sensor boundary.

Let $\mathcal{P}$ denote eligible represented inputs and $\mathcal{E}$ eligible existence inputs, including qualified absence censors. For the receiver's delivered Metropolis weights w_j , define

$$a_j = \frac{w_j \mathbf{1}[j \in \mathcal{P}]}{\sum_{k \in \mathcal{P}} w_k}, \quad q_j = \frac{w_j f_j \mathbf{1}[j \in \mathcal{E}]}{\sum_{k \in \mathcal{E}} w_k f_k}. \quad (2)$$

For a represented label with a positive timestamp set

$$f_j = \rho + (1 - \rho) \exp[-(t - \tau_j^\ell) \Delta t / \lambda]. \quad (3)$$

Use $f_j = \rho$ for a represented but never-directly-observed label, and $f_j = 1$ for a qualified absence censor. Here $\rho = 0.25$ and $\lambda = 5$ s are fixed before development outcomes. The age factor is independent of the sign of the existence claim: a recent low r_j can outweigh an older high r_j . Positive self weight and the existing absence logic supply the ordinary single-input fallback; a label without eligible spatial support is omitted.

C. Existence recency with fixed spatial weights

For clarity suppress the label and receiver indices. Denote Bernoulli relative entropy by D_B and ordinary density relative entropy by D . For fixed eligibility and normalized a, q , consider

$$\min_{r \in (0,1), p} J(r, p) = \sum_{j \in \mathcal{E}} q_j D_B(r \| r_j) + r \sum_{j \in \mathcal{P}} a_j D(p \| p_j). \quad (4)$$

When the two supports and weight vectors agree, this is the Bernoulli KLA objective. Allowing separate vectors changes that objective; ordinary KLA consistency claims cannot simply be transferred.

Proposition 1 (fixed-input form). Assume $0 < r_j < 1$, $\eta_a = \int \prod_{j \in \mathcal{P}} p_j(x)^{a_j} dx > 0$, and finite relevant divergences. A minimizer of (4) is

$$p^*(x) = \eta_a^{-1} \prod_{j \in \mathcal{P}} p_j(x)^{a_j}, \quad (5)$$

$$\text{logit } r^* = \sum_{j \in \mathcal{E}} q_j \text{logit } r_j + \log \eta_a. \quad (6)$$

For fixed a , eligibility, and input densities, changing q changes existence without changing p^* .

Proof. Writing $p_a = \eta_a^{-1} \prod_{j \in \mathcal{P}} p_j^{a_j}$ gives $\sum_j a_j D(p \| p_j) = D(p \| p_a) - \log \eta_a$. For every $r > 0$ this is minimized at $p = p_a$. Differentiating the remaining scalar objective yields

$\text{logit } r - \sum_j q_j \text{logit } r_j - \log \eta_a = 0$. Its second derivative is $1/[r(1-r)] > 0$, establishing the stated solution. $\square$

The overlap term remains a penalty: for exact normalized densities, $0 < \eta_a \leq 1$. Thus the rule does not guarantee cardinality consistency, probability calibration, or an existence value between all inputs. Nor does the proposition guarantee identical recursive localization: different existence decisions can affect subsequent associations, pruning, and which states are extracted. We test localization on common assignment support in addition to global set metrics.

The implementation calls the existing mixture-aware Gaussian-power fusion routine with a for spatial pooling and q for existence. It retains at most three components per input for the powered-mixture calculation, at most 256 component tuples, and at most eight output components. A non-admissible approximate overlap normalizer triggers the existing moment-matched fallback. Fractional powers of Gaussian mixtures and component reduction are approximate, so the proposition describes the ideal densities rather than an exact mixture implementation. All arms retain the same eight-component cap, prediction, measurement-update code, and MAP-cardinality extraction. We call the rule existence recency (ER); using q in both blocks is the distinct both-block recency ablation.

D. Confirmation ablation and communication cost

A separate ablation delays lineage restriction until some input carries a positive confirmation. A robot creates this bit after two consecutive local updates with opportunity and detection-association mass at least 0.5. Empty measurements reset the local streak, even if the update retains old diagnostic fields. Only the bit propagates; relays cannot supply consecutive local hits or refresh direct age. Ordinary FoV-aware pooling applies before confirmation. This conventional two-hit gate tests a false-positive versus short-contact discovery tradeoff; it is not presumed beneficial.

All communicating arms serialize the full retained mixture and identical metadata, then decode the packet before receiver fusion. Each attempted packet occupies 16,384 bytes, with a fixed 128-byte-per-robot control charge per frame. Padding isolates fusion at equal attempted wire cost. We also report unpadding payloads because differing mixture sizes can change actual data volume. The method adds constant metadata per represented label and linear-time weight calculations; it does not reduce the scheduled traffic.

IV. Experimental Protocol

A. Prescribed mobile sensing scenarios

Eight simulated robots follow two smooth patrol families for 120 frames at $\Delta t = 0.5$ s. Their maximum speed is below 3 m/s and acceleration below 2.1 m/s². Radio range is 24 m and circular sensing range is 14 m. The paths induce connected and two-component

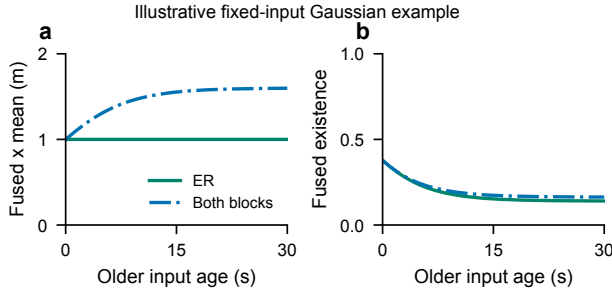


Fig. 2. Illustrative fixed-input example, not simulation results. Two unit-covariance Gaussian densities have x means 0 and 2 m, with existence 0.9 at the older source and 0.1 at the current source. Ordinary spatial weights are equal. Increasing the older source’s direct age changes ER’s existence while leaving its spatial mean at 1 m. Applying age to both blocks also moves the spatial mean. Both rules retain their respective spatial overlap normalizer.

periods; they do not respond to tracking outcomes. All targets use a constant-velocity state with independent per-step acceleration standard deviation 0.016 m/s^2 . The filter uses a continuous white-acceleration covariance with intensity 0.0128 and survival probability 0.995. Within FoV, detection probability is 0.9 and Cartesian observation noise has standard deviation 0.4 m. Poisson clutter has mean one per robot per frame and is uniform over a fixed $104 \times 80 \text{ m}$ rectangle. This deliberately simple sensor model does not include occlusion or robot localization error. Local pruning discards existence at or below 0.01; belief propagation uses tolerance 10^{-6} and at most 1,000 iterations. These settings are identical across arms.

The split-rejoin scene has two initial targets and two targets born at frame 35 while the teams are separated. The churn-departure scene uses repeated contact changes and removes one late target after frame 90; the filters receive no departure notification. In the no-new control, only the two initial targets exist, although identical later birth priors are still offered. Six fixed candidate regions have prior existence 0.1, position standard deviation 3.2 m, and velocity standard deviation 0.2 m/s. Regions 1–2 are activated at frame 1 and regions 3–6 at frame 35. Regions 5–6 remain empty in every scene. Actual initial offsets and target count are not given to the filter. Birth schedules and broad regions are known, so the task is not unrestricted open-world discovery.

Development uses seeds 2801–2803 and is explicitly separated from validation. Validation uses all seeds 2901–2920 in every family. Each robot receives an independent constant path offset uniform in $[-0.6, 0.6]^2 \text{ m}$, drawn before observations; physical components and trees are recomputed. Constant offsets preserve speed and acceleration and produce modest layout/contact variation. This is not twenty independently designed navigation environments. Truth, detections, clutter, and directed link-loss draws are paired across all arms. Directed packet loss is 0.1.

B. Controls and ablations

The no-communication Local arm provides context. FoV uses the existing observable-absence censor. Lineage adds untouched-prior exclusion; this is the main control for ER. Age-all applies direct-age weights to both spatial and existence pooling without lineage exclusion. L+Age-all additionally uses lineage exclusion, separating placement of the age factor from qualification. ER uses the same qualification with age in existence alone. Confirmed ER adds the two-hit gate.

The MIL-Z control uses constrained LMB MIL, $r = \sum_j w_j r_j$ and $p = \sum_j w_j r_j p_j / r$, with zero existence for absent labels [9]. MIL-S specializes label-subspace MIL [9], [10] to common label keys: for each source-membership pattern, renormalize weights on represented sources before the same formula. Its union of Gaussian components is reduced to the common cap. Untouched but represented priors participate. This support specialization does not reproduce the published independent-label assignment pipeline, and neither MIL arm is a new algorithm.

Every nonlocal arm uses the same attempted links, delivered links, weights before label processing, packet size, and control charge. The local update and extraction are unchanged. All pre/post estimates and input caches are saved. The original code and every experiment protocol are hash recorded.

C. Metrics and uncertainty

We report position OSPA with order 2 and cutoff 12 m [16], cardinality absolute error, and GOSPA with $p = 2, c = 12 \text{ m}, \alpha = 2$ [17]. Localization, missed-target, and false-target GOSPA components are reported as squared costs in m^2 . All extant targets are scored at every robot, including targets that are currently inaccessible. Thus a large error can represent a communication-limited discovery failure rather than poor local sensing.

For localization comparisons, assignments are computed separately for each method and matched errors are pooled only on robot–time–truth triples matched within the cutoff by both methods. The RMSE and its support must be read together; unmatched targets remain penalized by OSPA/GOSPA. Departure false cost uses frames 91–120. Reunion OSPA uses the next ten frames, truncated at a further disconnection. For opposite-subteam robots, acquisition is the first of three consecutive assignments within 2 m of a late target, censored at that target’s departure or the episode end. This measures geometric discovery, not label continuity. A separate Python evaluator exhaustively enumerates the small rectangular assignments and checks all saved MATLAB set metrics, independently of its Hungarian implementation. This is an independent metric implementation, not an independent research replication.

Uncertainty uses the twenty paired episodes as units, not the individual robots or frames. We report paired

mean differences and descriptive 95% percentile bootstrap intervals from 10,000 seed resamples. Intervals are not adjusted for multiple comparisons and are not presented as confirmatory significance tests. No validation outcome changes a method, threshold, birth prior, path, or included seed.

V. Results

A. Development results and the frozen decision

The initial Age-all arm passed a coarse direction screen against ordinary and FoV-aware KLA, but lost to the stronger Lineage control and worsened localization on common targets. Existence-only recency then reduced development OSPA against Lineage by 5.6% and 14.0% in the two event families, while retaining seven false estimates in 2,880 no-new robot-frames; FoV had none. The confirmation gate eliminated those extra false estimates but increased churn OSPA by 60.5% against Lineage. Neither the v2 nor v3 combination met its full balanced screen. We retained these failures, froze the rules, and ran the new-seed study to assess the conditional mechanism rather than searching for a passing threshold.

B. Existence recency and set estimation

Table I includes all 540 validation runs. Against Lineage, ER changes mean OSPA from 2.651 to 2.569 m in split-rejoin and from 1.123 to 0.846 m in churn-departure. The paired differences are -0.082 m [-0.110, -0.051] and -0.277 m [-0.399, -0.159], respectively. In the no-new control, the difference is -0.129 m [-0.167, -0.095]. Brackets denote the descriptive bootstrap intervals defined above. Cardinality error decreases in all three families. In the predeclared reunion windows, OSPA is 2.533 versus 3.264 m in split-rejoin and 0.926 versus 1.248 m in churn-departure (ER versus Lineage).

FoV and the two MIL variants have higher OSPA in these inputs. Support partitioning improves MIL in the event families but does not remove its localization and false-target tradeoffs. These results apply to the stated shared-label specialization and finite mixture cap, and do not establish superiority over complete published multi-view or label-matching systems. The large Local error reflects the deliberately global evaluation during separation; it should not be interpreted as a failed local detector.

C. Where the age factor belongs

Table II and Fig. 3(b) distinguish existence gains from localization. With the same lineage qualification, L+Age-all increases common-target RMSE relative to Lineage by 16.8% in split-rejoin and 3.1% in churn-departure. ER's ratios are 0.9999 and 0.9998, over 57,988 and 58,936 common triples. Those differences are negligible relative to measurement noise. This supports the fixed-input motivation in Fig. 2: recent direct sensor opportunity is useful for existence weighting, but does not measure the total spatial information retained in a posterior. The

recursive result is empirical and specific to these scenes; the proposition alone does not establish it.

D. False hypotheses and the confirmation tradeoff

The false-cost results materially limit the conclusion (Table III). In development ER lowered post-departure false cost by 34.5% against Lineage. In validation it instead increases the mean from 10.785 to 12.255 m². The paired difference is +1.470 m² [-0.705, 3.795]; the development improvement does not reproduce. Similarly, ER's no-new false cost is 0.570 m², versus 0.00375 m² for FoV. Lower OSPA therefore does not imply better false-track rejection.

Confirmed ER reduces no-new false cost to 0.00375 m², but its churn OSPA is 1.519 m, versus ER's 0.846 m. This is consistent with the cost of requiring consecutive local positive associations during short sensing opportunities. In churn, ER acquires all 160 opposite-subteam robot-target queries across twenty episodes; Confirmed ER acquires 152. Among acquired queries, mean delays are 2.47 and 7.63 s, respectively; the eight censored confirmed queries are excluded from the latter mean and remain failures. The gate is a disclosed ablation, not an accepted remedy. The unsmoothed trajectories in Fig. 4 show why discovery and post-departure persistence should be evaluated separately.

E. Accounting and verification

All communicating arms have exactly the same attempted and delivered messages for each paired episode. Mean attempted wire cost is 24.98 MiB for split/no-new and 25.74 MiB for churn. Across communicating arms and families, mean unpadded payload totals range from 2.14 to 9.85 MiB; all occupy the same padded slots. The metric audit recomputes 518,400 validation robot-frames and verifies 1,169 frozen source hashes. Focused checks verify signed age behavior, the no-opportunity fallback, source-order invariance, fixed-input spatial invariance, and exact v2 replay under the added confirmation metadata. These checks establish the stated implementation properties and agreement between the two metric implementations.

VI. Discussion and Conclusion

The useful result is conditional: direct-opportunity recency can improve existence decisions without also changing spatial weights. With a retained lineage qualifier, this improves global set estimation in the studied robot encounters and avoids the common-target localization degradation of both-block age weighting. It does not solve every failure induced by intermittent sensing. In particular, observation lineage can preserve an incorrect hypothesis, and a robot with old direct age may carry valuable relayed evidence. A confirmation gate can reject useful short-contact discoveries as well as false hypotheses. The failed development gate and reversed departure result make that limitation explicit.

TABLE I

Validation on all twenty paired seeds per family. OSPA is mean $\pm$ episode SD (m); count is mean absolute cardinality error. All nine arms and all seeds are included.

Method	Split-rejoin		Churn-departure		No-new control	
	OSPA	Count	OSPA	Count	OSPA	Count
Local	7.099 ± 0.454	1.368	7.280 ± 0.537	1.308	6.004 ± 0.839	0.645
FoV	3.713 ± 0.584	0.614	2.867 ± 0.551	0.442	0.804 ± 0.163	0.056
Lineage	2.651 ± 0.106	0.411	1.123 ± 0.421	0.143	0.797 ± 0.264	0.056
MIL-Z	4.812 ± 0.499	0.641	4.204 ± 0.574	0.452	2.765 ± 0.854	0.032
MIL-S	4.762 ± 0.506	0.624	4.112 ± 0.582	0.431	2.769 ± 0.854	0.033
Age-all	2.912 ± 0.123	0.466	1.960 ± 0.533	0.292	0.745 ± 0.172	0.043
L+Age-all	2.589 ± 0.090	0.399	0.937 ± 0.250	0.115	0.752 ± 0.233	0.045
ER	2.569 ± 0.110	0.398	0.846 ± 0.205	0.103	0.668 ± 0.219	0.040
Confirmed ER	2.607 ± 0.107	0.403	1.519 ± 0.440	0.220	0.652 ± 0.157	0.037

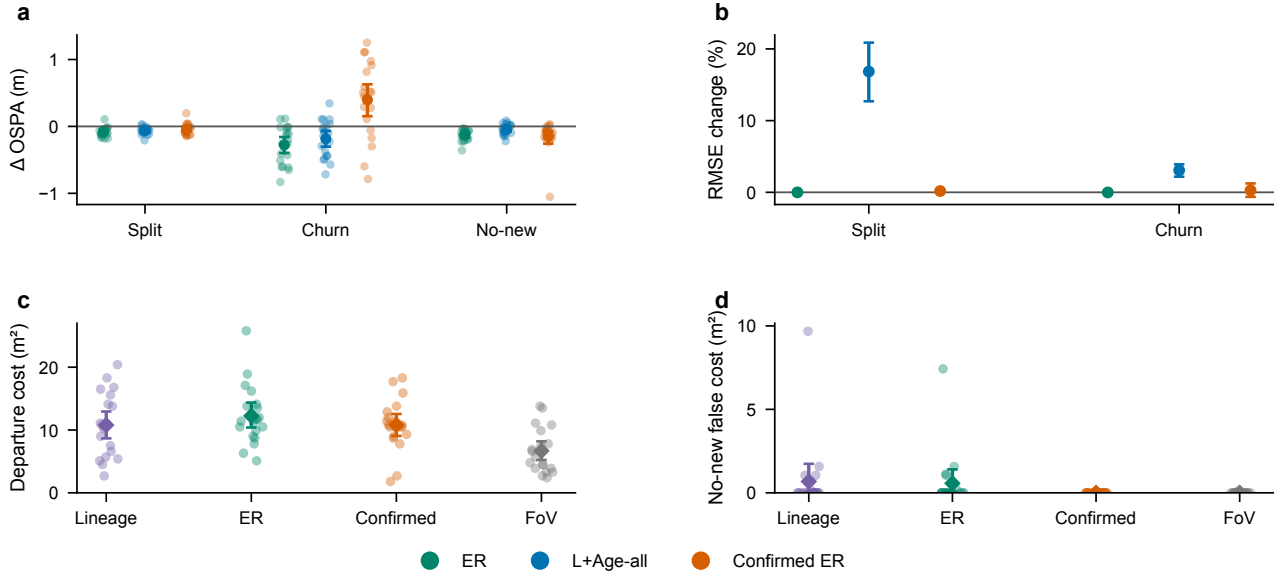


Fig. 3. Frozen validation outcomes. (a) Paired OSPA differences to Lineage; small dots are the twenty episode differences, and large points with bars show means and descriptive 95% bootstrap intervals. (b) Pooled RMSE ratios on common assignment support, displayed as percentage changes; intervals resample episodes. (c,d) Departure and no-new false-target squared costs, with twenty episode points per method. ER improves set error while preserving common-target localization, but does not eliminate false hypotheses. Confirmation has a different discovery/false-cost tradeoff.

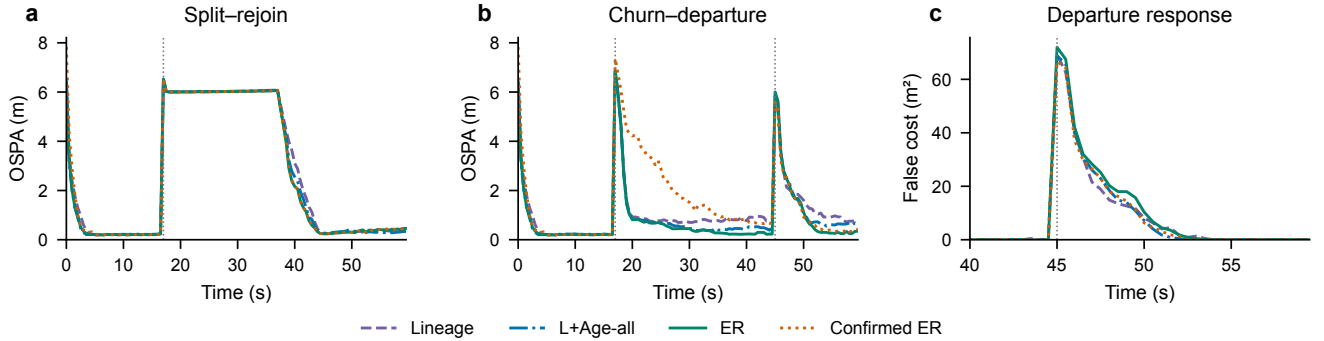


Fig. 4. Mean trajectories across the twenty validation episodes; no smoothing is applied. Dotted event markers indicate late birth at 17 s and, in churn, departure at 45 s. ER and the both-block ablation can improve set error while differing in localization and false-target persistence. The plotted means do not imply independent time samples; episode-level uncertainty is reported in Fig. 3.

TABLE II

Common-target localization relative to Lineage. Ratio is pooled candidate RMSE divided by reference RMSE; brackets are descriptive 95% bootstrap intervals. n_c counts common robot–time–truth triples, not independent samples.

Scene	Method	RMSE ratio [95% interval]	n_c
Split	ER	0.9999 [0.9997, 1.0001]	57,988
Split	L+Age-all	1.1685 [1.1270, 1.2087]	58,006
Split	Confirmed	1.0018 [1.0004, 1.0032]	57,894
Churn	ER	0.9998 [0.9997, 1.0000]	58,936
Churn	L+Age-all	1.0307 [1.0218, 1.0390]	58,841
Churn	Confirmed	1.0028 [0.9937, 1.0124]	56,641

TABLE III

Residual false-target GOSPA squared cost (m^2), averaged over twenty episodes. Departure uses only frames 91–120; no-new uses the entire episode.

Method	Departure	No-new
Local	10.935	0.611
FoV	6.675	0.004
Lineage	10.785	0.679
MIL-Z	23.430	4.162
MIL-S	26.370	4.204
Age-all	9.690	0.004
L+Age-all	10.905	0.570
ER	12.255	0.570
Confirmed ER	10.815	0.004

The main threats to generality are the small target population, shared birth schedule and label keys, known global poses, simplified Cartesian detections, mean-based FoV tests, fixed paths, one fusion round, and limited mixture representation. Unknown birth locations, independent-label association, occlusion, localization error, sensor heterogeneity, and real robot execution are not validated. The simulations also use a reliable central geometry coordinator with modeled control traffic. Neither network-protocol feasibility nor closed-loop navigation follows from them.

Within this scope, observation history should be treated as distinct from spatial information when designing a fusion rule for intermittently connected robot teams. The present evidence supports existence-only age weighting as a small fusion-layer change, with an explicit false-track tradeoff. Broader deployment requires evaluating that tradeoff under realistic birth, sensing, and label-association processes.

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